Lecture 15
Random Variables and Discrete Probability Distributions

Review

1 What is a sample space?

The collection of all possible outcomes of a random trial.

2 What is the law of large numbers?

As the number of trials grows, the relative frequency of A gets closer and closer to P(A).

3 What are disjoint (mutually exclusive) events?

Events with no outcomes in common, so they cannot happen together: $A \cap B = \emptyset$

Two more rules

general addition rule

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$: subtract the overlap so it is not counted twice

multiplication rule

if A and B are independent (one occurring does not change the probability of the other): $P(A \cap B) = P(A) \cdot P(B)$

disjoint ≠ independent

disjoint events cannot happen together, so knowing one happened tells you the other did not

Example: one die

Roll a fair six-sided die once.

S = {1, 2, 3, 4, 5, 6}, each outcome has probability 1/6

(a) What is P(not a six)?
(b) What is P(1 ∪ 6)?
(c) What is P(even ∪ greater than 3)?

(a) 1 – 1/6 = 5/6 (rule 4)
(b) 1/6 + 1/6 = 1/3 (rule 3: a 1 and a 6 are disjoint)
(c) 3/6 + 3/6 – 2/6 = 4/6 (the 4 and the 6 are in both)

Example: two dice

Roll a fair die twice. The two rolls are independent.

P(six) = 1/6 on each roll

(a) What is P(six on both rolls)?
(b) What is P(at least one six)?

(a) (1/6)(1/6) = 1/36 (multiplication rule)
(b) 1 – P(no sixes) = 1 – (5/6)(5/6) = 11/36 ≈ 0.31

Data: student heights

A survey of 379 college students recorded each student's height and gender.

X = the height of a student, in inches

HEIGHT	GENDER
56	Female
57	Female
58	Female
60	Female
60	Female
…	…

n = 379 students (the file lists the women first, then the men, each sorted by height)

Example: unions and intersections

Pick one of the 379 students at random. F = female, T = at least 70 in.

	At least 70 in	Under 70 in	Total
Female	19	243	262
Male	87	30	117
Total	106	273	379

(a) What is P(F ∩ T)?

(b) What is P(F ∪ T)?

(a) 19 / 379 ≈ 0.05

(b) 262/379 + 106/379 – 19/379 = 349/379 ≈ 0.92

Are F and T independent?

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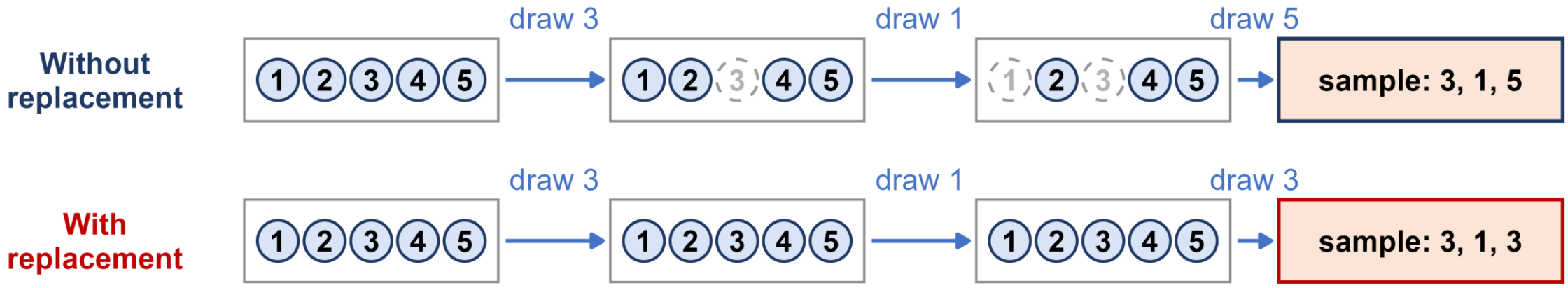
Does knowing a student is female change the chance they are at least 70 in?

Yes: 106/379 ≈ 28% of all students, but only 19/262 ≈ 7% of the women

Not independent: $P(F) \cdot P(T) \approx 0.69 \cdot 0.28 \approx 0.19$, but $P(F \cap T) \approx 0.05$

With or without replacement?

Week 4: drawing a sample of 3 balls from a bag of 5, two ways.



Are the draws independent?

With replacement: yes. The bag is the same on every draw, so $P(A \cap B) = P(A) \cdot P(B)$
Without replacement: no. Each draw changes what is left, so the next probability depends on the last draw

Example: drawing from an urn

An urn holds 3 red, 4 blue and 3 yellow balls. Draw 2 balls without replacement.

10 balls in all: 3 red, 4 blue, 3 yellow

(a) What is P(both red)?
(b) What is P(no red balls)?
(c) What is P(both the same color)?

(a) (3/10)(2/9) = 6/90 ≈ 0.067
(b) (7/10)(6/9) = 42/90 ≈ 0.467
(c) (3/10)(2/9) + (4/10)(3/9) + (3/10)(2/9) = 24/90 ≈ 0.267

Example: drawing two aces

A standard deck has 52 cards, 4 of them aces. Draw two cards.

P(ace on the first draw) = 4/52

(a) What is P(two aces) if the first card is put back (with replacement)?
(b) What is P(two aces) if it is not put back (without replacement)?

(a) (4/52)(4/52) = 1/169 ≈ 0.0059: the draws are independent
(b) (4/52)(3/51) = 1/221 ≈ 0.0045: only 3 aces are left among 51 cards

Random variables

random variable

the outcome of a random trial recorded as a number, written with a capital letter such as X

discrete

its possible values can be listed: the number of heads in 3 tosses, X = 0, 1, 2, 3

continuous

it can take any value in an interval: a student's height

Example: three coin tosses

Toss a fair coin 3 times: 2 × 2 × 2 = 8 equally likely outcomes.

X = the number of heads

(a) List the sample space S.

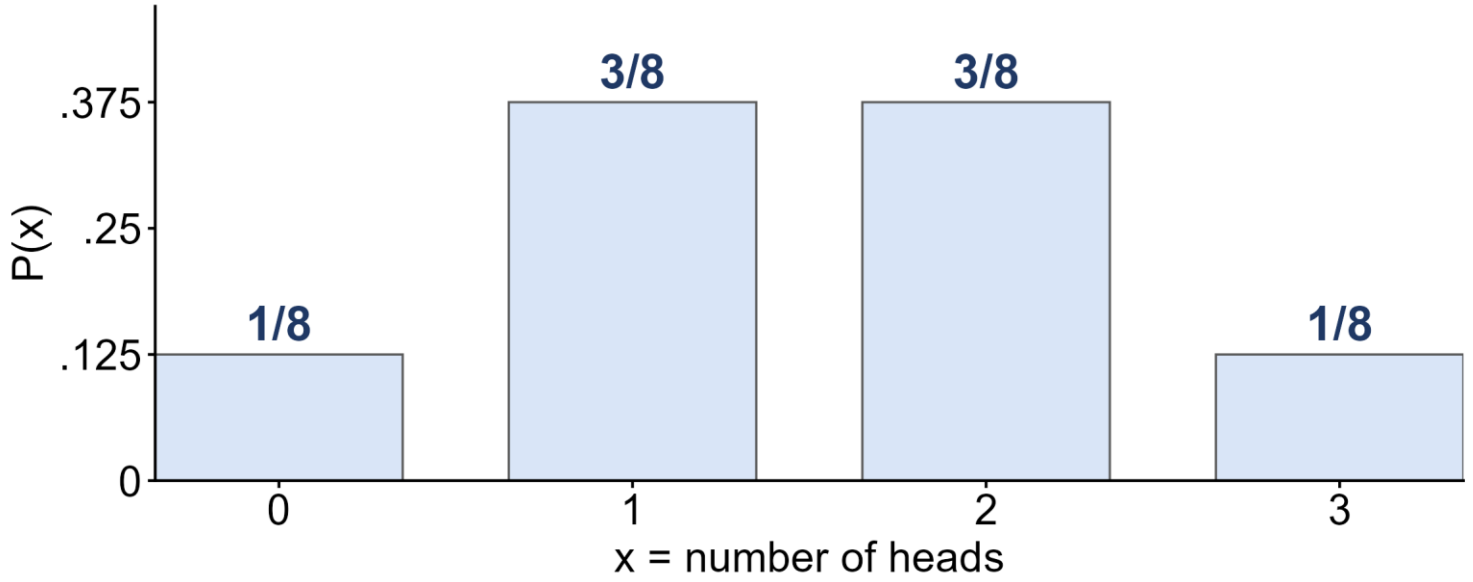
(b) How many outcomes give each value of X?

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

X = 0: 1 X = 1: 3 X = 2: 3 X = 3: 1 (out of 8)

Discrete probability distributions

probability distribution a function that assigns a probability to each possible value of X; each is between 0 and 1, and they add to 1



1 + 3 + 3 + 1 = 8, so the probabilities add to 8/8 = 1

Example: using a distribution

X = the number of heads in 3 tosses of a fair coin.

P(0) = 1/8 P(1) = 3/8 P(2) = 3/8 P(3) = 1/8

(a) What is P(X ≥ 2)?
(b) What is P(X ≤ 1)?
(c) What is P(1 ≤ X ≤ 2)?

(a) 3/8 + 1/8 = 4/8
(b) 1 – P(X ≥ 2) = 1 – 4/8 = 4/8
(c) 3/8 + 3/8 = 6/8

The mean of a random variable

expected value μ

the long-run average of X: $\mu = \Sigma\, x \cdot P(x)$

variance σ^2

$\sigma^2 = \Sigma\, (x - \mu)^2 \cdot P(x)$, and the standard deviation is $\sigma = \sqrt{\sigma^2}$

three coin tosses

$\mu = 0(1/8) + 1(3/8) + 2(3/8) + 3(1/8) = 1.5$ heads

Example: grade points

A STAT 251 exam grade on the 4-point scale, for a student chosen at random:

x	0	1	2	3	4
grade	F	D	C	B	A
P(x)	0.05	0.10	0.30	0.25	0.30

(a) What is P(better than a C)?

(b) What is the mean grade, μ?

(a) P(X = 3) + P(X = 4) = 0.25 + 0.30 = 0.55

(b) μ = 0(0.05) + 1(0.10) + 2(0.30) + 3(0.25) + 4(0.30) = 2.65

Bernoulli trials

Bernoulli trial

one trial with two outcomes: success (X = 1) with probability p, failure (X = 0) with probability 1 − p

X ~ Bernoulli(p)

its single parameter p is the probability of success

mean and variance

$\mu = p$ and $\sigma^2 = p(1 - p)$

Example: guessing one question

You guess on a multiple-choice question with 4 options, one of them correct.

X = 1 if the guess is right, X = 0 if not; p = 1/4 = 0.25

(a) What is P(X = 0)?

(b) What are μ and σ^2?

(a) $1 - 0.25 = 0.75$

(b) $\mu = 0.25$, $\sigma^2 = 0.25 \times 0.75 = 0.1875$

The binomial setting

a fixed number of trials n

the trial is repeated a set number of times, decided in advance

two outcomes, the same p

each trial is a success or a failure, with the same probability of success p every time

independent trials

the outcome of one trial does not change the probability of success on any other

The binomial distribution

$X \sim \text{Binom}(n, p)$

X = the number of successes in n independent trials, each with probability of success p

$P(X = k)$

$n! / (k!(n - k)!) \cdot p^k \cdot (1 - p)^{n-k}$: the first factor counts the ways to arrange k successes

mean and variance

$\mu = np$ and $\sigma^2 = np(1 - p)$

Example: three tosses again

Friday: X = the number of heads in 3 tosses of a fair coin, counted from the 8 outcomes.

X ~ Binom(n = 3, p = 0.5)

(a) Use the formula to find P(X = 2).

(b) What does 3! / (2! 1!) count?

(a) $3!/(2!\,1!) \cdot 0.5^2 \cdot 0.5^1 = 3 \times 1/8 = 3/8$: the same answer as counting

(b) the 3 orders with 2 heads: HHT, HTH, THH

Example: guessing a whole quiz

You guess on all 10 questions of a 4-option multiple-choice quiz.

$X \sim \text{Binom}(n = 10, p = 0.25)$

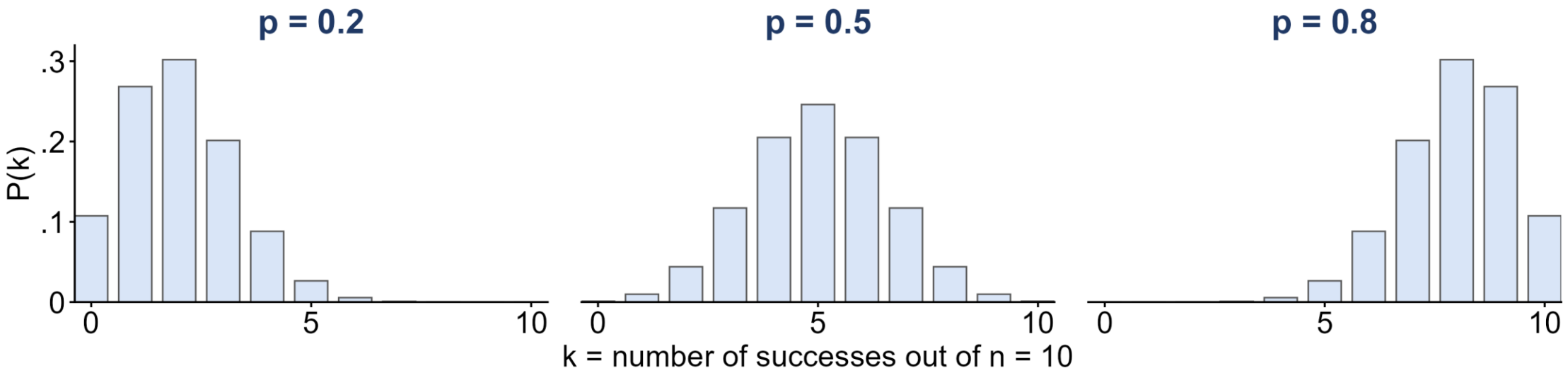
(a) What is $P(X = 3)$?
(b) What is $P(X = 0)$?
(c) What are μ and σ?

(a) $10!/(3!\,7!) \cdot 0.25^3 \cdot 0.75^7 = 120 \times 0.0156 \times 0.1335 \approx 0.250$
(b) $0.75^{10} \approx 0.056$
(c) $\mu = 10 \times 0.25 = 2.5$, $\sigma = \sqrt{(10 \times 0.25 \times 0.75)} \approx 1.37$

The shape of a binomial distribution

X ~ Binom(n = 10, p) for three values of p.

What does p do to the shape?



p = 0.5: symmetric. p < 0.5: skewed right. p > 0.5: skewed left.
The center sits at μ = np: 2, 5 and 8

Example: free throws

Stephen Curry has made 4,236 of the 4,645 free throws he has attempted in the NBA.

p = 4,236 / 4,645 ≈ 0.912; he shoots n = 10; X = free throws made

(a) What is P(he makes all 10)?
(b) What is P(he misses at least one)?
(c) How many does he make, on average?

(a) $0.912^{10} \approx 0.40$
(b) $1 - 0.40 = 0.60$
(c) $\mu = np = 10 \times 0.912 \approx 9.1$

Is it binomial?

For each X, check the three conditions: fixed n, the same p, independent trials.

(a) X = the number of heads in 20 flips of a coin
(b) X = the number of aces in a 5-card hand
(c) X = the number of rolls of a die until the first six

(a) Yes: n = 20, p = 0.5, and the flips are independent
(b) No: the cards are dealt without replacement, so the draws are not independent
(c) No: the number of trials n is not fixed

The Poisson distribution

X ~ Poisson(λ)

X = the number of events in a fixed interval of time or space, when events occur independently at an average rate λ

P(X = k)

$\lambda^k e^{-\lambda} / k!$, k = 0, 1, 2, … (e ≈ 2.718)

mean and variance

$\mu = \lambda$ and $\sigma^2 = \lambda$

Example: World Cup goals

The 2022 FIFA World Cup had 172 goals in 64 matches.

$\lambda = 172 / 64 \approx 2.69$ goals per match; X = goals in one match

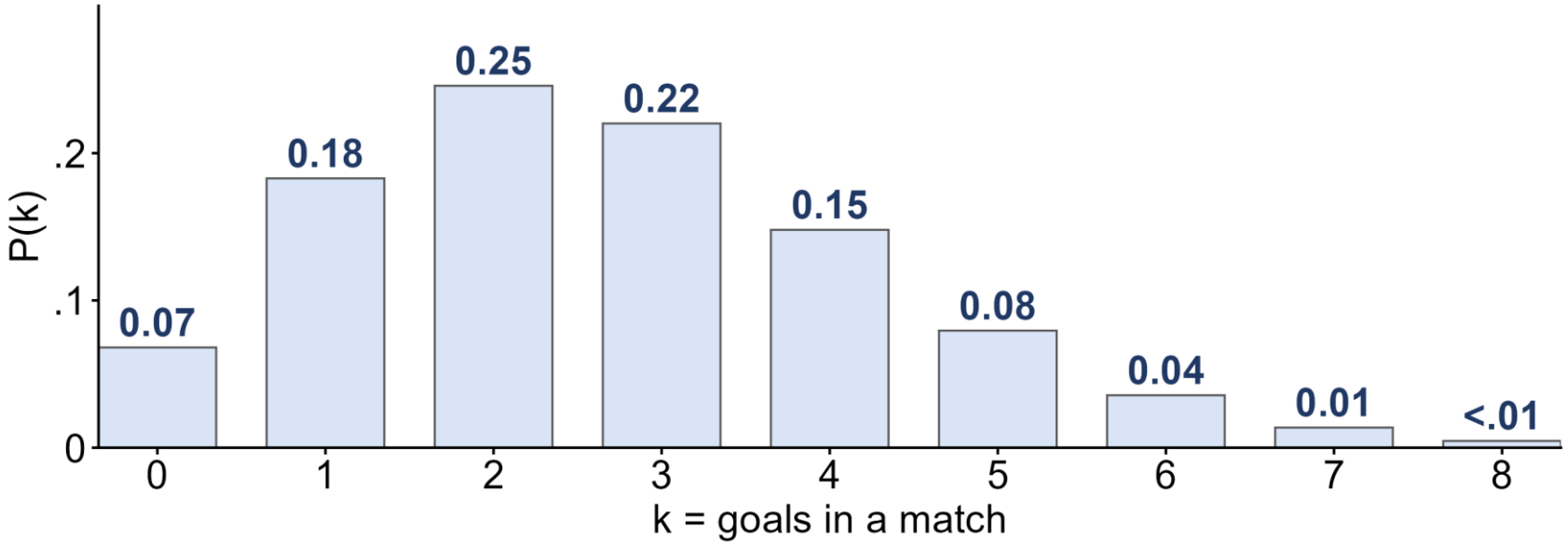
(a) What is $P(X = 0)$?
(b) How many of the 64 matches does the model expect to end 0–0? (There were 7.)
(c) What is λ for one 45-minute half, and P(no goals in a half)?

(a) $e^{-2.69} \approx 0.068$
(b) $64 \times 0.068 \approx 4.4$: close, but a model is never a perfect fit
(c) $\lambda = 2.69 / 2 \approx 1.34$, $P(X = 0) = e^{-1.34} \approx 0.26$

Goals per match: the Poisson model

X ~ Poisson(λ = 2.69): the probability of each number of goals in a match.

Which number of goals is most likely? Is the distribution symmetric?



Two goals (0.25), then three (0.22); skewed right: X cannot go below 0 but has no upper limit

What you should know: discrete distributions

probability rules

$P(A') = 1 - P(A)$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; independent: $P(A \cap B) = P(A)\,P(B)$

random variables

a discrete distribution lists each value of X with its probability, adding to 1; $\mu = \Sigma\, x\, P(x)$

binomial

successes in n independent trials with the same p: $\mu = np$, $\sigma^2 = np(1 - p)$; Bernoulli is $n = 1$

Poisson

events in a fixed interval at an average rate λ: $\mu = \sigma^2 = \lambda$